Lesson 2.1: EDA with Python, feature engineering and predictive models

What you learn

- **EDA.** Look at shape, types, missing values, distributions and relationships using Pandas, Matplotlib and Seaborn.
- **Feature engineering.** Create useful inputs such as margin percentage or days since last purchase.
- **Predictive models.** Train on a split and measure on unseen data; see the machine learning lesson in Department 2.

Tools and resources (official sites, not checked live)

- Seaborn
- Matplotlib

ASSIGNMENT

Practice for lesson 2.1

Business problem: Create a feature.

Steps

1. Run the EDA code above.
2. Add a new feature column.
3. Explain why it may help a model.

Expected output: A new column and a reason.

SOLUTION (ANSWER KEY)

Margin percentage is a feature because it combines revenue and cost into one comparable number.

Check yourself

1. What is a feature?
   Answer: An input column for a model.
2. What does EDA reveal?
   Answer: Data quality and patterns.

Lesson 2.1 of the **Data Analytics and Data Science** lessons. See this lesson and its official links on skillnavora.in · All chapters and lessons · Open the full lesson guide. Tools, prices and plans change; confirm on each official site.